Pivotum

THE DEGREE RISK INDEX · FALL 2026 EDITION

Is a Biology Degree Worth It in the AI Era?

The complete profile — the full scores by track, the six factors behind them, the honest downsides, and what to ask before you commit.

5.0

AI EXPOSURE · CLINICAL TRIALS / FIELD RESEARCH

where 10 is most at risk

Read this one first; send the student version to them.

HIGHLIGHTING IS COLOUR-CODED

THE FINDING EXPOSURE PROTECTION METHOD & HONESTY

Before you read this — a note for parents

You're likely reading this before your child is. That's normal, and it's worth pausing on how you use it.

Don't lead with the scores. If you open with a risk number, they either get frightened and shut down, or defensive about a decision they've already made.

What this is. A facts document, not a recommendation.

A practical note: there's a separate short version written directly to them. Send them that one.

And one thing specific to life sciences. The AI finding here is genuinely surprising and the financial finding is genuinely important, and they're different things.

The surprise: the computational, tech-forward end of biology is *more* exposed than the hands-on laboratory end. The important part: a biology bachelor's on its own leads to technician work at around $47,000, while the same field in industry averages three times that. Section 6 is the one to read carefully.

The short answer

Yes for hands-on and regulated work. Much less so for the computational end — which is the opposite of what almost everyone assumes.

Field ecology and environmental science scores **4.2** out of 10. Wet lab and bench research scores **5.2.** Bioinformatics and computational biology scores **7.1.**

The people at the bench are better protected than the people analysing what the bench produces.

And the bigger question for most students isn't exposure at all — it's that a life sciences bachelor's is a gateway qualification rather than a destination, and where it leads varies by a factor of three in pay.

The scores

TRACK	2023	2025	FALL 2026	3-YR MOVEMENT	BAND
Field ecology & environmental science	3.6	3.9	**4.2**	+0.6	Low–Moderate
Clinical trials & regulatory affairs	3.9	4.3	**4.6**	+0.7	Moderate
Wet lab & bench research	4.4	4.8	**5.2**	+0.8	Moderate
Lab technician & diagnostics	4.9	5.4	**5.8**	+0.9	Moderate–High
Bioinformatics & computational biology	5.6	6.4	**7.1**	+1.5	High
Scientific data curation & literature analysis	6.4	7.2	**7.8**	+1.4	High

Scores run 1–10, where 10 is most at risk. For context: the median profession in this edition sits around 5.5, entry data analysis scores 8.2, licensed engineering 4.0, and bedside nursing 2.8.

Read the table this way, and note the ordering. The most technically sophisticated, best-paid and most competitive track in this list is also the most exposed. That inversion is the whole profile.

The inversion: why computational biology is more exposed than the bench

This is the finding worth carrying beyond biology, because the same pattern appears in agriculture, construction and several other fields in this index.

The intuition most people have is that the tech-forward version of a career must be the future-proof one. Bioinformatics is computational, quantitative, well-paid and hard to get into. Wet lab work is manual, repetitive and lower-status. Surely the first is safer.

The framework says the opposite, and the reason is structural.

Ask what each job actually consists of.

Bioinformatics takes structured data — sequences, expression matrices, imaging output — and produces analysis and code. Structured inputs, computational outputs, no physical presence, no licensed accountability. That is the exposed profile in every field we score, and it doesn't stop being the exposed profile because the subject matter is biology.

Wet lab work requires hands. Pipetting, cell culture, animal handling, sample preparation, running an instrument that has developed a fault nobody documented. It happens in a physical space, on physical material, and the thing that goes wrong is usually not in any protocol.

So computational biology is exposed for the same reason data analysis is exposed — because it is data analysis. The biology is the subject; the shape of the work is computation.

The general lesson: proximity to technology is not protection. We make this point in the computer science and data science profiles too, and life sciences is the clearest demonstration because the two tracks sit inside one degree. Same students, same department, opposite exposure.

One important qualification. This is a statement about the *analysis* layer, not about scientists. A computational biologist who designs the experiment, decides what question is worth asking and is accountable for the interpretation is doing protected work. One who runs pipelines on other people's data is not. The distinction is judgment and ownership, exactly as everywhere else in this index.

Why the bench holds — the six factors

How much of this job can AI already do?	**6.5**
How hard will it be to land that first job?	**6.0**
Does it have to be done in person, with your hands?	**7.5**
Does someone need a human they can trust and hold responsible?	**5.0**
Does the law require a licensed human?	**3.0**
How often does the job hit genuinely new, high-stakes situations?	**8.0**

EXPOSURE RAISES THE SCORE   PROTECTION LOWERS IT

Here's how wet lab and bench research answers them.

HOW MUCH OF THIS JOB CAN AI ALREADY DO? (35% OF THE SCORE)

The analysis, the literature work and the writing. Not the experiment.

Sequence and structural analysis. Statistical analysis and visualisation. Literature review and synthesis. Protocol drafting. Manuscript preparation. Image quantification.

And genuinely: hypothesis generation from existing data is now meaningfully assisted, which is a real change to how research is done.

What resists: doing the experiment. Noticing that a culture looks wrong before any measurement says so. Troubleshooting an instrument. Handling animals. And the tacit craft that makes one person's results reproducible and another's not.

HOW HARD WILL IT BE TO LAND THAT FIRST JOB? (15%)

Moderate, and this is where the honest problems sit — not with exposure but with pay and progression. Section 6 covers it.

DOES IT HAVE TO BE DONE IN PERSON, WITH YOUR HANDS? (15%)

Substantially, for bench and field work. Not at all for computational work. That single difference does most of the work in the score table.

DOES SOMEONE NEED A HUMAN THEY CAN TRUST AND HOLD RESPONSIBLE? (15%)

In regulated environments, significantly. Clinical trials, GMP manufacturing and diagnostics all require named responsible individuals, and that accountability is a real protection. In academic research it is weaker.

DOES THE LAW REQUIRE A LICENSED HUMAN? (10%)

In regulated work, yes — and this is under-appreciated. Clinical trial conduct, GMP manufacturing and clinical diagnostics operate under regulatory frameworks requiring qualified persons. In basic research, essentially none.

HOW OFTEN DOES THE JOB HIT GENUINELY NEW, HIGH-STAKES SITUATIONS? (10%)

Constantly in research — novelty is the entire point of the work — though the stakes of any individual experiment are usually low.

The financial picture, and it matters more than the score

This is the section that should shape the decision.

A life sciences bachelor's on its own leads to technician work.

- Biological technicians earn a median of around **$47,000–$48,000**
- Entry-level lab technician and research assistant roles typically run **$40,000–$55,000**
- Clinical research coordinators earn a median of about **$52,880**
- Environmental scientists do better, at a median near **$80,000**
- Forensic science technicians earn around **$63,740**, with 11% projected growth

Now compare the same field in industry.

National average base salaries for life sciences professionals in industry sit in the **mid $140,000s to low $160,000s**. Process engineers and manufacturing science specialists earn **$110,000–$160,000.** Senior quality and compliance professionals earn **$90,000–$140,000**, with heads of quality exceeding $180,000. Commercial roles start at **$110,000–$170,000** base.

That is a gap of roughly three times between where a biology degree most commonly lands and where the same field pays best.

The gap is not about intelligence or ability. It's about sector. Academic and bench-adjacent roles are funded by grants and university budgets. Industry roles are funded by products. The science overlaps considerably; the economics do not.

And the growth numbers are respectable rather than exciting: medical scientists around 9%, environmental scientists 8%, biological technicians 7%, biochemists and biophysicists 6%, postsecondary biology teachers 5%. Life sciences overall around 5% through 2032, with biotech adding over 20,000 positions in 2025.

Two practical conclusions follow.

The degree is a gateway, not a destination. Most people who do well from a life sciences bachelor's do something else afterwards — medicine, dentistry, veterinary, physician associate, a specialist master's, or a move into industry, regulatory affairs or commercial roles.

And the PhD question needs answering honestly. Doctoral qualifications show a 40–50% salary uplift over bachelor's in BLS classifications, at the cost of four to seven years. That is a genuine return in industry and a much weaker one in academia, where the number of permanent research positions is far smaller than the number of people trained for them.

What the trend shows

Wet lab and bench research: 4.4 → 4.8 → 5.2. Up 0.8 in three years.

Bioinformatics: 5.6 → 6.4 → 7.1. Up 1.5 — nearly double the pace.

The gap has inverted further. In 2023 the difference was 1.2 points; it's now 1.9. The computational end is not converging with the bench end. It's separating from it, in the direction nobody expected.

How durable is the protection?

PROTECTION	STRENGTH TODAY	DURABILITY	WHERE THE PRESSURE IS COMING FROM
Physical / embodied	Real for bench and field work	**Durable for now**	Lab automation is real but expensive and inflexible; unstructured troubleshooting resists.
Regulatory / licensing	Strong in regulated environments	**Most durable**	Clinical trials, GMP and diagnostics require named responsible persons.
Judgment under novelty	Strong in research	**Eroding at the analysis layer**	Hypothesis generation is now assisted; experimental design and interpretation less so.
Human trust / accountability	Real in regulated roles	**Durable**	Someone signs off on a batch, a trial, a diagnostic result.

The honest read. The protected version of a life sciences career is the regulated one: clinical trials, quality, manufacturing, diagnostics. That is also where the pay is. The two findings point the same way, which is unusual and worth noticing.

How the role will actually change

WAS THE JOB	IS BECOMING THE JOB
Running the analysis pipeline	Directing it, then interrogating the output
Literature review	Synthesised automatically; the scientist judges relevance
Drafting the manuscript	Edited rather than written
Generating hypotheses from data	Increasingly assisted — a genuine change
Doing the experiment	**Unchanged**
Noticing the culture looks wrong	**Unchanged**
Signing off a regulated batch or trial	**Unchanged**

One development worth naming honestly. Hypothesis generation from existing data is the first genuinely intellectual part of science to be meaningfully assisted, and the profession has not settled what that means. Designing the right experiment and interpreting an unexpected result remain human. Proposing candidate explanations increasingly is not.

Human strengths that will matter most

1. **Doing the experiment well** — the tacit craft that makes results reproducible. The least

> automatable and least teachable thing here.

2. **Noticing something is wrong** before any measurement says so. 3. **Regulated accountability** — being the named person who signs off a batch, a trial or a

> diagnostic result.

4. **Experimental design** — deciding what would actually answer the question. 5. **Interpreting the unexpected** — the result that doesn't fit, which is where most real

> discovery lives.

Notably absent: analysis pipeline fluency, literature recall and figure production. These were traditional markers of a strong graduate researcher and are the fastest-depreciating.

Other things to consider about this job

What's genuinely good about it

The work is genuinely interesting, and people mean it. Research is one of very few careers where the point is to find out something nobody knew.

- **Growth is steady across most specialisms** — 9% for medical scientists, 8% for environmental scientists, 11% for forensic technicians
- **Industry pay is strong** — mid $140,000s to low $160,000s average base, with manufacturing, quality and commercial roles all well above the academic equivalent
- **Biotech added over 20,000 positions in 2025**
- **The degree opens an unusually wide set of doors** — medicine, dentistry, veterinary, physician associate, public health, and industry
- **Environmental and field roles** combine science with outdoor work and score well at 4.2

And regulated work is the under-considered good answer. Regulatory affairs, quality assurance and clinical trial management pay well, are protected by named accountability, and almost nobody applies to them from a biology degree.

What's genuinely hard about it

The bachelor's alone leads to modestly-paid technician work, and this is the dominant practical problem. Median around $47,000, with limited progression without further qualification.

The academic pipeline is the honest difficulty. Far more people are trained for research careers than there are permanent research positions, and the postdoctoral stage can extend for years on short contracts. Anyone considering an academic path should understand the numbers before committing four to seven years.

Funding is grant-dependent in academia, which makes job security genuinely uncertain in a way industry roles are not.

And the entry-level market is competitive. Life sciences graduates are numerous, and the roles their degree points at directly are neither plentiful nor well paid.

Who this work suits — and who it doesn't

Life sciences tends to suit people who:

- **Are genuinely curious about how living things work** — the subject has to be interesting on its own terms, because the early pay isn't compensating
- **Are patient and precise** — most experiments fail, and reproducibility is a craft
- **Can tolerate uncertainty** over long periods
- **Are practically dextrous** if aiming at bench work — this is manual skill and it varies
- **Are willing to plan the next step** — the degree is a gateway

It's harder for people who:

- Want **strong early earnings** — the technician tier is where most graduates start
- Need **certainty and closure** — research rarely provides either
- Assume the **degree is the destination.** This is the most common and most costly mistake in this profile.

What else is moving this market

Biotech and pharmaceutical hiring is the growth engine, and it is where the pay is.

Regulatory complexity is expanding, which creates demand for people who understand both the science and the compliance framework — and there are not many.

Lab automation is real but uneven — high-throughput, standardised work automates readily; bespoke and troubleshooting-heavy work does not.

And the academic funding environment is the volatile part, which affects the research career path far more than any technology.

The AI-native advantage — what to actually do about it

The situation: the analysis layer of biology is automating quickly, and the experimental and regulated layers are not. That points to a clear strategy.

WHAT AN AI-NATIVE LIFE SCIENTIST LOOKS LIKE

1. Own the question, not the pipeline. Running an analysis is no longer a skill. Deciding what analysis would answer the question, and whether the answer is believable, is.

2. Keep the hands. Bench and field capability is the physical protection, and it is what computational-only colleagues lack.

3. Move toward regulated work. Clinical trials, GMP manufacturing, quality and diagnostics all require named accountable humans — and they pay considerably better than academic research.

4. Verify. Knowing when a generated analysis or a synthesised literature summary is confidently wrong requires understanding the underlying biology well enough to smell it.

5. Understand hypothesis assistance honestly. These tools genuinely help generate candidate explanations. The scientist's value is increasingly in designing the discriminating experiment and interpreting what comes back.

CONCRETE PREPARATION

Before the degree: understand that this is a gateway qualification and have a working theory of what comes next — medicine, a specialist master's, industry, or regulated work.

During: get real lab or field experience early and often; it is the differentiator and it is what employers actually assess. Take statistics seriously — not as a hurdle but as the judgment layer. And look at industry placements rather than only academic ones.

Choosing a direction: if the pull is computational, pair it with genuine experimental capability rather than treating it as the alternative to lab work.

The routes in

ROUTE	ASSESSMENT
Biology / biomedical bachelor's → professional programme	Medicine, dentistry, veterinary, PA, pharmacy. A very common and successful destination.
Bachelor's → industry (QC, regulatory, manufacturing)	Under-applied to, protected by named accountability, and pays far better than the technician tier.
Bachelor's → specialist master's	Regulatory affairs, clinical research, biotechnology, public health.
PhD → industry research	40–50% salary uplift, four to seven years. Genuine return in industry.
PhD → academic career	The honest one: far more trained than there are permanent positions.
Environmental / field science	Scores best at 4.2, median near $80,000, 8% growth.

Where a life sciences degree can lead later

Genuinely broad: clinical and professional healthcare routes, industry research and development, manufacturing and process science, quality and regulatory affairs, clinical trial management, diagnostics, environmental and conservation science, forensic science, scientific communication, policy, teaching, and commercial roles in biotech and pharmaceuticals.

The pattern in this index holds with an unusual twist. The further from physical work *and* from named accountability, the higher the exposure. Pure analysis roles score worst precisely because they have neither.

What to look for in a life sciences program

1. Lab and field time — how much, and how early? This is the differentiator, and programmes vary enormously.

2. Statistics and experimental design, taught as reasoning. Not as a service module. This is the judgment layer that survives.

3. Industry placement, not only academic research placement. Given the pay gap, this matters a great deal.

4. Does the curriculum address regulated environments? GMP, GLP, clinical trial regulation. Rarely taught and highly employable.

5. First-destination outcomes by sector, not overall employment rate.

Questions to ask a life sciences program

On experience:

- How many hours of hands-on lab or field work, and starting in which year?
- Are industry placements available, and what proportion of students get one?

On the curriculum:

- How much statistics and experimental design, and is it compulsory?
- Does the programme cover regulated environments — GMP, GLP, clinical trials?
- How does it address AI-assisted analysis, and where do students learn to verify it?

On outcomes:

- Where are graduates at one year — technician roles, industry, professional programmes, or further study?
- What proportion enter industry rather than academia?

Red flags: minimal lab time before final year; statistics as an optional module; no industry placement route; outcomes described only as "employed or in further study."

Go deeper — further reading

- **Biological Technicians — Occupational Outlook Handbook** — *US Bureau of Labor Statistics*. The pay and growth reality for the role most biology bachelor's graduates enter first. Read the **Medical Scientists** and **Environmental Scientists** pages alongside it — the contrast is the useful part.

- **BLS Life, Physical, and Social Science Occupations** — the full set, with growth projections and median pay for every track in this profile. Free and comparable.

- **US Life Sciences Salary Guide 2026** — industry compensation by function: manufacturing, quality, regulatory, commercial. Note it's published by a staffing firm, so read it for the shape of the market rather than as neutral — but the gap it shows against academic pay is the important finding.

The counterargument — read this one:

> The strongest case against our 7.1 for bioinformatics is that we are scoring the *technique* rather than the *science*.
>
> On this view, a computational biologist is not a data analyst who happens to work on biology — they are a scientist whose instrument is a computer. The judgment about which analysis is meaningful, which result is artefactual and which finding is worth pursuing is scientific judgment, and it is no more automatable than a bench scientist's. Automating the pipeline no more replaces the computational biologist than automating a sequencer replaced the molecular biologist.
>
> **Where we land:** we think this is right about the best computational biologists and wrong about the median role. Our score reflects the work as most commonly performed — running established pipelines on data someone else generated, to answer questions someone else framed. The scientists who design and interpret are protected. The distinction is judgment and ownership, and it happens to be exactly the distinction that separates tracks in every other profile in this index.
>
> **What we're watching:** whether computational roles increasingly carry experimental design responsibility. If they do, the score should come down.

Bottom line

Life sciences contains an inversion worth understanding: the computational end is more exposed than the bench.

Bioinformatics scores 7.1 against wet lab research at 5.2 — and the gap is widening rather than closing. The reason is structural: computational biology takes structured inputs and produces analysis, which is the exposed profile in every field we score. The biology is the subject; the shape of the work is computation.

But the more important finding is financial. A life sciences bachelor's most commonly leads to technician work at around **$47,000**, while the same field in industry averages **mid $140,000s to low $160,000s** — a gap of roughly three times, driven by sector rather than by ability.

So the useful advice is: treat the degree as a gateway and have a working theory of the next step before enrolling. Keep hands-on capability rather than trading it for computational skill. Aim at regulated work — clinical trials, quality, manufacturing, diagnostics — because it is both the protected version and the well-paid one, and almost nobody applies to it from a biology degree. Take statistics seriously as judgment rather than as a hurdle. And be honest about the academic path, where far more people are trained than there are permanent positions.

And one thing worth saying to a student who simply finds living things fascinating. That's a good reason to study this, and it's not sufficient on its own. The people who do well from a life sciences degree are almost always the ones who worked out early what it was a gateway to.

Discussion questions

Part 1 — About the work itself

1. What made biology appealing? Push past "I like science" to something specific. 2. **Do they want to do experiments, or analyse data?** They're different careers with a

> 1.9-point exposure gap inside one degree.

3. Are they practically dextrous? Bench work is manual skill and it varies more than people

> expect.

4. How do they handle things not working? Most experiments fail.

Part 2 — Reacting to the findings

5. Bioinformatics scores 7.1 and wet lab 5.2. Was that inversion surprising? 6. The profile argues proximity to technology isn't protection. Can they apply that idea to

> another career?

7. **Hypothesis generation from data is now meaningfully assisted.** What do they make of that?

Part 2b — The money conversation

8. **Biological technicians earn a median around $47,000; industry life sciences averages mid

> $140,000s to low $160,000s.** Have you both looked at that gap squarely?

9. Do they have a working theory of what the degree is a gateway to? This is the single

> most important question in this profile.

10. If the plan is a PhD, have you looked at how many permanent academic positions exist relative

> to the number of people trained?

Part 2c — The other side

11. Of what's rewarding — finding out something nobody knew, breadth of options, field work,

> industry pay — which two matter most?

12. Looking at "who this work suits": which items sound like them? Answer separately, then

> compare.

Part 2d — The AI question, practically

13. Running an analysis is no longer a skill; deciding what analysis would answer the question

> is. How would they get good at the second thing?

14. Have they considered pairing computational skill with genuine experimental capability rather

> than treating them as alternatives?

Part 3 — Testing it against reality

15. **The most valuable single action:** get real lab or field experience before committing. It's

> the differentiator on applications and it answers the fit question.

16. Find someone three to five years past a biology degree and ask what they actually do and what

> they earn.

Part 4 — About the program

17. Ask each programme the lab-hours and industry-placement questions from section 14. 18. Is statistics compulsory? Does the curriculum cover regulated environments?

Part 5 — Widening the frame

19. **Have they looked at regulatory affairs, quality assurance or clinical trial management?**

> Protected, well paid, and almost nobody applies from a biology degree.

20. What about **environmental and field science**? Best score in this profile at 4.2, median near

> $80,000, 8% growth.

21. If the real pull is healthcare, have they compared this against **nursing, physician

> associate or allied health** — all of which reach licensed practice faster?

Part 6 — For the parent, alone

22. Do I know what this degree actually leads to, or am I assuming "science" means secure? 23. Am I comfortable with a gateway qualification that needs a second step? 24. What's the one thing I most want them to understand — in a sentence, without a number in it?

This is analysis and informed judgment about a fast-moving situation. For life sciences specifically, we've flagged that our score reflects the median computational role rather than the best ones, and that the profession's most significant practical issue — the gap between academic and industry compensation — sits entirely outside what our six factors measure.

Scores for 2023 and 2025 are retrospective reconstructions using the current methodology. Scores measure exposure to what AI can already do — not how much any particular employer has deployed.

Technical scoring appendix — Life Sciences

Pivotum Degree Risk Index — Fall 2026 Edition

This appendix is the audit trail. It sets out the formula, the rating anchors, every factor rating behind every track score in this profile, the backcast arithmetic, a sensitivity analysis, and the specific things that would change our mind.

We publish it because a score nobody can check is an opinion with a decimal point.

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1. The formula

```
Risk = 0.35 × Automatability
     + 0.15 × EntryErosion
     + 0.15 × (10 − Physical)
     + 0.15 × (10 − Trust)
     + 0.10 × (10 − Regulatory)
     + 0.10 × (10 − Judgment)
```

Two exposure factors carry 50% of the weight; three protection factors carry 40%; judgment under novelty carries the remaining 10% and sits on the protection side. The deliberate consequence is that protection can outweigh exposure — which is why a highly automatable job like teaching still scores low.

Bands. Very low below 3.0 · Low 3.0–4.4 · Moderate 4.5–5.4 · Moderate–High 5.5–6.9 · High 7.0 and above.

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2. Rating anchors

HOW THE 0–10 RAW RATINGS ARE ANCHORED

Every factor is rated on its own 0–10 scale before weighting. The anchors are identical across all twenty-seven careers, which is what makes the scores comparable.

Task automatability (A) — *what share of the working week could a capable current system already do to an acceptable standard?*

RATING	ANCHOR
0–2	Almost nothing. The work is physical, relational or wholly novel.
3–4	Administrative surround only — notes, scheduling, correspondence.
5–6	A meaningful minority of the week, mostly documentation and lookup.
7–8	A large share, including tasks central to the junior version of the role.
9–10	Most of the described duties, at or near professional standard.

Entry-path erosion (E) — *how much has the traditional route into this work narrowed?*

RATING	ANCHOR
0–2	Protected by statute or physical necessity. Training hours cannot be automated.
3–4	Stable. Some junior tasks absorbed; the apprenticeship survives.
5–6	Visible narrowing. Employers prefer experience; junior intake reduced.
7–8	Substantial. The work juniors learned on has largely gone.
9–10	The training layer and the automatable layer were the same layer.

Physical / embodied (P) — *must this be done in person, with a body, in an unpredictable setting?*

RATING	ANCHOR
0–2	Fully screen-based. Location-independent.
3–4	Occasional presence; the core work is not physical.
5–6	Regular physical presence, in largely controlled conditions.
7–8	Substantially physical, with some variability of setting.
9–10	Irreducibly physical, in conditions that differ every time.

Human trust / accountability (T) — *does someone need a specific human they can rely on, and must a human answer when it goes wrong?*

RATING	ANCHOR
0–2	Output is anonymous and reviewed by others.
3–4	Work is signed off by someone else. No personal reliance.
5–6	Some client relationship; accountability is institutional.
7–8	Named responsibility, with commercial or professional consequence.
9–10	The relationship is the service, and liability attaches personally.

Regulatory / licensing (R) — *does the law reserve this act to a qualified human*?

RATING	ANCHOR
0–2	No licence exists and none is proposed.
3–4	Certification is customary but not reserved.
5–6	Partial reservation, varying by jurisdiction or task.
7–8	Licensed practice, with some unreserved adjacent work.
9–10	Comprehensive statutory reservation, including of the training route.

Judgment under novelty (J) — *how often does the work present something with no matching precedent*, *where being wrong is costly*?

RATING	ANCHOR
0–2	Routine by design. Deviation is an error, not a case.
3–4	Occasional exceptions, handled by escalation.
5–6	Regular non-standard cases within a known range.
7–8	Frequent genuine novelty with material consequences.
9–10	Novelty is constant, and in some fields adversarially generated.

3. Factor ratings by track

Ratings are the Fall 2026 edition unless a range is shown. **P, T, R and J are held constant across editions** — those protections are structural and do not move on a three-year horizon. All measured movement is in A and E.

FIELD ECOLOGY & ENVIRONMENTAL SCIENCE — 4.2 (LOW)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	5.0	5.7	**6.4**	2.24
Entry-path erosion	15%	3.0	3.4	**3.8**	0.57
Physical / embodied	15%	8.5	8.5	**8.5**	0.22
Human trust / accountability	15%	6.5	6.5	**6.5**	0.53
Regulatory / licensing	10%	5.5	5.5	**5.5**	0.45
Judgment under novelty	10%	8.0	8.0	**8.0**	0.2
				Total	**4.21 → 4.2**

CLINICAL TRIALS & REGULATORY AFFAIRS — 4.6 (MODERATE)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	5.2	6.2	**6.9**	2.42
Entry-path erosion	15%	3.2	3.6	**4.0**	0.6
Physical / embodied	15%	3.5	3.5	**3.5**	0.97
Human trust / accountability	15%	8.5	8.5	**8.5**	0.22
Regulatory / licensing	10%	9.0	9.0	**9.0**	0.1
Judgment under novelty	10%	7.0	7.0	**7.0**	0.3
				Total	**4.61 → 4.6**

WET LAB & BENCH RESEARCH — 5.2 (MODERATE)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	5.7	6.6	**7.6**	2.66
Entry-path erosion	15%	3.5	4.0	**4.4**	0.66
Physical / embodied	15%	8.0	8.0	**8.0**	0.3
Human trust / accountability	15%	5.5	5.5	**5.5**	0.67
Regulatory / licensing	10%	2.5	2.5	**2.5**	0.75
Judgment under novelty	10%	8.5	8.5	**8.5**	0.15
				Total	**5.2 → 5.2**

LAB TECHNICIAN & DIAGNOSTICS — 5.8 (MODERATE–HIGH)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	6.7	7.9	**8.9**	3.11
Entry-path erosion	15%	4.0	4.5	**5.0**	0.75
Physical / embodied	15%	7.5	7.5	**7.5**	0.38
Human trust / accountability	15%	5.5	5.5	**5.5**	0.67
Regulatory / licensing	10%	5.5	5.5	**5.5**	0.45
Judgment under novelty	10%	5.5	5.5	**5.5**	0.45
				Total	**5.82 → 5.8**

BIOINFORMATICS & COMPUTATIONAL BIOLOGY — 7.1 (HIGH)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	4.9	6.8	**8.6**	3.01
Entry-path erosion	15%	4.5	5.2	**5.8**	0.87
Physical / embodied	15%	1.0	1.0	**1.0**	1.35
Human trust / accountability	15%	5.5	5.5	**5.5**	0.67
Regulatory / licensing	10%	1.0	1.0	**1.0**	0.9
Judgment under novelty	10%	7.0	7.0	**7.0**	0.3
				Total	**7.1 → 7.1**

SCIENTIFIC DATA CURATION & LITERATURE ANALYSIS — 7.8 (HIGH)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	5.7	7.7	**9.2**	3.22
Entry-path erosion	15%	5.5	6.2	**6.8**	1.02
Physical / embodied	15%	1.0	1.0	**1.0**	1.35
Human trust / accountability	15%	4.5	4.5	**4.5**	0.82
Regulatory / licensing	10%	1.0	1.0	**1.0**	0.9
Judgment under novelty	10%	5.0	5.0	**5.0**	0.5
				Total	**7.82 → 7.8**

4. Backcast arithmetic

Worked example for the headline track, showing exactly how the three-year movement is composed.

Field ecology & environmental science

EDITION	A	E	PROTECTION DEFICIT (FIXED)	SCORE
2023	5.0	3.0	1.4	**3.6**
2025	5.7	3.4	1.4	**3.9**
Fall 2026	6.4	3.8	1.4	**4.2**

Movement decomposition: automatability contributed **+0.49** and entry-path erosion **+0.12**, for a total of **+0.6** points over three years.

The 2023 and 2025 figures are retrospective reconstructions using the current methodology, not archived scores from published editions. They are our best assessment of what each factor would have rated at the time, given what was then demonstrable. They are not measurements.

5. Sensitivity

How far the score moves if a single factor is rated one point differently:

FACTOR	±1 POINT MOVES THE SCORE BY
Task automatability	±0.35
Entry-path erosion	±0.15
Physical / embodied	±0.15
Human trust / accountability	±0.15
Regulatory / licensing	±0.10
Judgment under novelty	±0.10

What this means in practice. A one-point disagreement about automatability moves a score by 0.35 — enough to shift a track between bands at the margins. A one-point disagreement about licensing moves it by 0.10, which almost never changes a band.

So if you disagree with one of our numbers, the automatability rating is where the disagreement matters most, and it is the rating we would most want challenged.

6. Stated limitations

We measure capability, not deployment. Scores reflect exposure to what AI can already do at the leading edge — not how much any particular employer has adopted. Adoption lags capability by years and lags unevenly: large, well-funded organisations first; small, rural, public-sector and thin-margin employers much later. The lag is a buffer, not a shelter. It buys time without changing direction.

We do not measure demand. A task can be highly automatable and highly in demand simultaneously. Where the labour market and our framework disagree, we say so in the profile rather than adjusting the score to fit.

We do not measure oversupply. The number of people entering a field is outside our six factors, and for some careers it matters more than automation.

We do not measure industry economics. A profession can contract for reasons entirely unrelated to technology.

We do not measure whether the work is worth doing. Pay, satisfaction, debt, hours and meaning sit outside the score and are covered separately in each profile — often at greater length, because for several careers they matter more.

Sub-track definitions are ours. Job titles vary between employers and countries. We have chosen tracks that reflect genuinely different work, not official classifications.

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7. What would change our mind

Each edition names specific, checkable indicators. If these move, the scores move — including downward.

Across the index:

- **Graduate hiring share.** If employers in the professions where we rate entry-path erosion highly return to hiring juniors at pre-2023 rates, those ratings fall.
- **Content of mandated training hours.** Several professions protect their on-ramp through required supervised experience. If the content of those hours shifts from producing work to supervising it, the protection weakens without any rule changing.
- **Offsite and prefabricated construction share.** The clearest test of our physical-protection ratings: automation performs well in controlled settings, so moving work indoors raises exposure without any robotics breakthrough.

- **Verified deployment rather than capability.** If adoption stalls materially below capability for several editions, our leading-indicator approach is overstating near-term risk and we will say so.

We will report movement in either direction. A framework that only ever revises upward is not measuring anything.

Scores measure exposure to what AI can already do — not how much any particular employer has deployed. The 2023 and 2025 figures are retrospective reconstructions using the current methodology.